

# Memory Matters

One-on-One with Scott Thirlwell from Corsair Memory on the state of DRAM and SSDs in India and globally

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Corsair is a popular name when it comes to DRAM modules and SSDs. We had a free-wheeling conversation with Scott Thirlwell, Director of Sales APAC – Corsair Memory where we discussed the current state of the SSD market, the testing procedure of DRAM modules, among other things. Read on...

**Majority of the Indian consumers are quite clueless when it comes to DRAM and generally go by what the counter guy is suggesting. How do you plan to educate the Indian consumers about DRAM?**

Our main aim is to educate our channel partners and retailers about DRAM. We want to make them aware of our offerings and which type of memory is targeted at which segment. We have carried out roadshows where we involve retailers and train them about the various types of Corsair DRAM modules and whom they are meant for. Ultimately they are the ones who influence consumers on what memory to purchase, if the consumers are not well informed. So we make sure that we highlight strong aspects of our offering to the retailers.

For consumers, we are not directly training them. But we have experts who are present on various technology forums, who are there to answer queries.

**How do you perceive your SSD market in India?**

Frankly speaking India being a price conscious market, we still are not seeing numbers which we are seeing in say Europe or China, but they are slowly picking up. With SSDs we manage to sell anywhere between 700 to 900 units/month which is in line with our expectations. In the last 9 months prices have fallen considerably and in the next couple



Scott Thirlwell, Director of Sales - APAC Corsair Memory

of months, we are expecting SSD prices to fall even further, which will see a boost in that sales number.

**Why do you think that the SSD prices will drop?**

Well the NAND chip manufacturing technology is improving. So now we will be able to pack in more NAND chips per wafer than what we were doing before. Due to this the volumes will increase and this will lead to a definite reduction in prices.

**What kind of testing process does a DRAM module go through?**

It is quite fascinating actually to see how the memory chips are selected. NAND memory chips are produced on a wafer. The wafer has to be graded for 1333 MHz and has to be graded based on the farthest chip. This grading may vary as we approach the centre of the wafer, which may be capable of higher speeds. The chip maker will not cut off individual chips from the wafer and grade them, entire wafers have to be graded. So the chip maker can claim stuff like 'the worst performing chip on this wafer is 1333 MHz'.

You can't take the chips at the centre, run them through tests and say that if they are capable of going to 2400 MHz, then the entire wafer will have all the chips going to those speeds. In some other wafer the chips in the centre may only be capable of going upto 1600 MHz. These speeds may vary from wafer to wafer. So we at Corsair, have to check each individual chip.

So after the chips have been cut out from the wafer, we pass them through the functionality test.

Each chip is carefully sorted and then passed through the tests. We have world class equipment that runs stress tests on these memory chips. So for instance if we have to include a chip in a 2133MHz RAM module, then that chip has to undergo stress tests for every speed rating starting from 1333MHz onwards.

When you observe a RAM stick, you will notice a number of memory chips on it, so while testing the RAM, we have a stick that has existing chips rated for the timing for which the chip has to be tested. In it you place the chip you want to test. If it succeeds that test, then it is tested for a higher timing. And so on. So if a chip does not clear the test for the 2133MHz module, it is used in the module which has slightly slower speeds, leaving apart some headroom for slight bit of overclocking as well.

**How do you decide on which controller to choose for the SSD?**

Testing. What we do is we take into consideration all the SSD controllers out there and our R&D team will test out each controller with the NAND memory that we have. Based on the results we decide upon which ones to use and which ones to discard.

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